

SAI NIKHIL RAVI

@ raavinikhil2212@gmail.com +1(856)312-3234 in linkedin.com/in/nikhil-r13011999 github.com/nikhil-raavi United States

SUMMARY

Data Engineer with hands-on experience building scalable ETL pipelines, optimizing distributed data processing, and enabling cloud-based analytics. Skilled in Python, PySpark, SQL, and modern data platforms such as AWS, Databricks, and Snowflake. Known for delivering reliable solutions with strong testing, automation, and collaboration.

EXPERIENCE

- Data Engineer, Epeople Technologies Inc (Client-Fiserv), United States

Jul 2024 – Jan 2025
- Developed and maintained **ETL pipelines** to ingest and transform large-scale customer, transactional, and financial data from multiple sources via **JDBC**, ensuring clean and reliable output for downstream analytics
 - Performed complex data transformations using **Spark SQL** and **PySpark**, including joins, aggregations, and window functions. Applied data cleansing techniques to handle nulls, filter invalid records, and create unified Delta Lake datasets for reporting.
 - Implemented advanced **Spark optimization** techniques like salting, broadcast joins, and caching in **Databricks**. Reduced job execution time from **4 hrs → 2.4 hrs** and resolved failure issues to ensure efficient data processing.
 - Optimized PySpark jobs in Azure Databricks by **tuning cluster** configurations using **dynamic allocation**, partition sizing improving throughput and reducing cloud **costs by 15%**
 - Led the incremental migration from an **on-prem Hadoop** system to Databricks, implementing autoscaling **clusters (10-50 nodes)** and tuning Spark configurations for efficient **resource utilization**
 - Established a data reconciliation process between **Snowflake** and **Greenplum** using **MD5 hashing**, ensuring data accuracy and automating discrepancy alerts for real-time monitoring
 - Migrated Spark ETL jobs from Hadoop HDFS to Cloud Databricks to snowflake, setting up Databricks and Snowflake.
 - Orchestrated ETL workflows using **Apache Airflow**, designing **dynamic DAG's** to automate data ingestion and transformation tasks, improving pipeline reliability by **12%** and reducing manual intervention
 - Developed automated ETL pipelines in **Databricks** to extract, transform, and load data from multiple sources into **S3**. Ensured seamless integration with AWS Databricks for scalable data processing and analytics.
 - Utilized **AWS EMR** to efficiently transform and transfer large volumes of data to and from various AWS data stores and databases, including S3 and **Amazon Redshift**
 - Designed and implemented a Snowflake data warehouse, configuring virtual warehouses (**compute clusters**) for scalable performance and cost optimization
 - Set up Amazon S3 as a staging area and leveraged Snowflake's COPY INTO command to efficiently load and transform raw data into Snowflake tables, ensuring high-performance ETL processing
- Programmer Analyst, Cognizant Technology Solutions, India

Dec 2019- Aug 2022
- Contributed to migrating **10+ legacy Teradata ETL** workflows to PySpark on **Databricks**, enabling **Sanofi's** analytics team to process **1M+** daily clinical trial records **30% faster** through parallelized data transformations
 - Converted **Teradata SQL** transformations, including patient cohort aggregations, into PySpark with row-level and **metric validation** to ensure accuracy in clinical trial data processing
 - Assisted **data analysts** in debugging issues related to missing, null, or incorrectly formatted clinical trial data by tracing root causes in ETL logic
 - Optimized complex SQL queries using window functions and indexing, reducing **query latency by 20%**
 - Monitored daily ETL pipelines in Databricks, assisting in resolving **10+ monthly job failures** and ensuring timely data availability for clinical dashboards
 - Wrote unit tests with **Pytest**, achieving **95% test** coverage and reducing post-deployment bugs
 - Supported ingestion of semi-structured clinical event data from **Kafka** into Databricks, transforming records with PySpark and validating outputs for ingestion into **Datawarehouse**
 - Performed daily validation checks to compare Teradata and Spark outputs using row counts and key column sums for **data accuracy**
- Software intern, A4 Softech, India

Jul 2019- Nov 2019
- Automated** Python scripts to extract and clean customer and transaction data of around **50k+ records** from SQL databases for reporting use cases
 - Helped optimize slow-performing queries by adding indexing and restructuring joins for faster access
 - Collaborated with the backend team to expose cleaned data via simple **REST APIs** for internal dashboards

SKILLS

Databricks

PySpark

Spark

Python

SQL

ETL

Snowflake

Informatica

AWS S3

Azure Data Factory

ADLS

Database

Datawarehouse

Data Pipeline

Airflow

Kafka

Data modeling

HDFS

Yarn

GIT

CI/CD

Redshift

PowerBI

Glue

Java

Hadoop

Hive

EDUCATION

Master of Science in Computer Science

Binghamton University, NY

📅 Sep 2022- May 2024

Bachelor of Engineering in Computer Science

Amrita Vishwa Vidyapeetham, India

📅 Jun 2016- May 2020

CERTIFICATIONS & ACHIEVEMENTS

- Databricks Certified Associate Developer for Apache Spark
- Awarded with Certificate of Achievement in PowerBI Course
- Certification of Achievement in Python by Dataflair
- Certification of Achievement in Snowflake by Snowflake
- SQL and Relational Database 101 Certified in collaboration with IBM

ACADEMIC PROJECTS

- Social Media Data Science Pipeline

Dec 2023
- Engineered a real-time pipeline using **Azure Event Hubs** to ingest **3K+ daily** posts from **Twitter, YouTube, and Reddit APIs**; stored raw data in **Azure Data Lake Storage** with lifecycle management policies, reducing storage costs by **30%**
 - Scheduled daily pipeline runs and cleanup tasks using Databricks notebooks and job triggers to simulate production workflows.
 - Optimized PySpark jobs in Azure Databricks to cleanse and transform **110GB+** unstructured data into structured formats, improving downstream analytics efficiency by **25%** through partition optimization and caching
 - Architected a star schema in Azure **Synapse Analytics** with distribution-optimized tables, enabling real-time Power BI dashboards to track **social sentiment trends (likes, shares, engagement)**